Question ID: c602140f

Question

$(x-11y)(2x-3y)-12y(-2x+3y)$

Which of the following is equivalent to the expression above?

Answer

A. $x-23y$

B. $2x^2-xy-3y^2$

C. $2x^2+24xy+36y^2$

D. $2x^2-49xy+69y^2$

Question ID: 0b3d25c5

Question

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x>0$?

Answer

A. $(x+4)^4$

B. $(x+4)^2$

C. $(x+4)$

D. $(x+4)^{\frac{1}{2}}$

Question ID: 8838a672

Question

$(4x^3-5x^2+3)-(6x^3+2x^2-x)$

Which of the following expressions is equivalent to the expression above?

Answer

A. $-10x^3-3x^2+x+3$

B. $-2x^3-7x^2+x+3$

C. $-2x^3-3x^2+x+3$

D. $10x^3-7x^2-x+3$

Question ID: b47419f4

Question

$$\left(\frac{1}{2}x+3\right)-\left(\frac{2}{3}x-5\right)$$

Which of the following is equivalent to the expression above?

Answer

A. $-\frac{1}{6}x+8$

B. $-\frac{1}{6}x-2$

C. $-\frac{1}{3}x^2+\frac{1}{2}x+15$

D. $-\frac{1}{3}x^2-\frac{9}{2}x-15$

Question ID: 7348f046

Question

$(2x+3)-(x-7)$

Which of the following is equivalent to the given expression?

Answer

A. $x-4$

B. $3x-4$

C. $x+10$

D. $2x^2+21$

Question ID: 3e9cc0c2

Question

Which of the following is equivalent to $(1-p)(1+p+p^2+p^3+p^4+p^5+p^6)$?

Answer

A. $1-p^8$

B. $1-p^7$

C. $1-p^6$

D. $1-p^5$

Question ID: d9137a84

Question

Which expression represents the product of $\left(x^{-6}y^{3}z^{5}\right)$ and $\left(x^{4}z^{5}+y^{8}z^{-7}\right)$?

Answer

A. $x^{-2}z^{10}+y^{11}z^{-2}$

B. $x^{-2}z^{10}+x^{-6}z^{-2}$

C. $x^{-2}y^{3}z^{10}+y^{8}z^{-7}$

D. $x^{-2}y^{3}z^{10}+x^{-6}y^{11}z^{-2}$

Question ID: 16de54c7

Question

$2x^2+5x-12$

If the given expression is rewritten in the form $(2x-3)(x+k)$, where k is a constant, what is the value of k ?

Question ID: c3a72da5

Question

Which of the following is equivalent to the sum of $3x^4+2x^3$ and $4x^4+7x^3$?

Answer

A. $16x^{14}$

B. $7x^8+9x^6$

C. $12x^4+14x^3$

D. $7x^4+9x^3$

Question ID: a391ed22

Question

$$\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$$

The expression above is equivalent to ax^2+bx+c, where a, b, and c are constants. What is the value of b?

Question ID: f237ccfc

Question

The sum of $-2x^2+x+31$ and $3x^2+7x-8$ can be written in the form ax^2+bx+c, where a, b, and c are constants. What is the value of $a+b+c$?

Question ID: 26eb61c1

Question

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

Answer

A. $6x^2y^2(2x^6)$

B. $6x^2y^2(x^4)$

C. $6x^2y^2(x^6 + 2)$

D. $6x^2y^2(x^4 + 2)$

Question ID: 6d04c89d

Question

The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where b is a constant and $x > 0$. What is the value of b?

Answer

A. 7

B. 10

C. 24

D. 252

Question ID: a1bf1c4e

Question

$x^2 + 6x + 4$

Which of the following is equivalent to the expression above?

Answer

A. $(x + 3)^2 + 5$

B. $(x + 3)^2 - 5$

C. $(x - 3)^2 + 5$

D. $(x - 3)^2 - 5$

Question ID: 463eec13

Question

If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$?

Answer

A. $8x^2y^4$

B. $4xy^4$

C. $4x^{-2}y^2$

D. $4x^{-1}y^4$

Question ID: ad2ec615

Question

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

Answer

A. $(x^2+1)(x^2-6)$

B. $(x^2+2)(x^2-3)$

C. $(x^2+3)(x^2-2)$

D. $(x^2+6)(x^2-1)$

Question ID: b8caaf84

Question

If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

Answer

A. $3x^2 + 12x + 7$

B. $3x^2 + 14x + 17$

C. $3x^2 + 19x + 20$

D. $3x^2 + 26x + 33$

Question ID: dd4ab4c4

Question

$4a^2+20ab+25b^2$

Which of the following is a factor of the polynomial above?

Answer

A. $a+b$

B. $2a+5b$

C. $4a+5b$

D. $4a+25b$

Question ID: 42c71eb5

Question

$(2x+5)^2-(x-2)+2(x+3)$

Which of the following is equivalent to the expression above?

Answer

A. $4x^2+21x+33$

B. $4x^2+21x+29$

C. $4x^2+x+29$

D. $4x^2+x+33$

Question ID: a05bd3a4

Question

Which of the following expressions is equivalent to x^2-5 ?

Answer

A. $(x+\sqrt{5})^2$

B. $(x-\sqrt{5})^2$

C. $(x+\sqrt{5})(x-\sqrt{5})$

D. $(x+5)(x-1)$

Question ID: cc776a04

Question

Which of the following is an equivalent form of $(1.5x-2.4)^2-(5.2x^2-6.4)$?

Answer

A. $-2.2x^2+1.6$

B. $-2.2x^2+11.2$

C. $-2.95x^2-7.2x+12.16$

D. $-2.95x^2-7.2x+0.64$

Question ID: a520ba07

Question

$\sqrt[3]{x^3y^6}$

Which of the following expressions is equivalent to the expression above?

Answer

A. y^2

B. xy^2

C. y^3

D. xy^3

Question ID: a255ae72

Question

If $x^2 = a + b$ and $y^2 = a + c$, which of the following is equal to $(x^2 - y^2)^2$?

Answer

A. $a^2 - 2ac + c^2$

B. $b^2 - 2bc + c^2$

C. $4a^2 - 4abc + c^2$

D. $4a^2 - 2abc + b^2c^2$